

AEROSPACE SUPPLY CHAIN RISK INTELLIGENCE

NAICS 336411 - Aircraft Manufacturing

MEDIUM

COMPOSITE RISK

46.43

GEO RISK

60.42

WORKFORCE RISK

32.44

Executive Summary

Aircraft Manufacturing exhibits Medium composite risk (46.43/100) driven primarily by trade concentration: 60% of \$10.0B in exports flow to a single destination country. Despite moderate workforce challenges (32.44/100), the segment's export dependency on one trading partner creates material procurement vulnerability.

Risk Drivers

- **Export destination concentration** - 60.42/100 risk score indicates approximately 60% of the segment's \$10.0 billion annual exports depend on a single foreign customer nation, exposing U.S. prime contractors to demand shocks from policy shifts, currency volatility, or geopolitical friction in that market.
- **Workforce availability pressure** - with 240,989 employed workers and a 32.44/100 workforce risk score, the segment faces moderate but persistent talent pipeline constraints, particularly for specialized roles in composite fabrication, avionics integration, and FAA-certified assembly.
- **Scale-dependent resilience** - \$10.0B export base and nearly 241,000 workers suggest deep industrial capacity, yet the Medium composite score (46.43/100) signals that size alone does not mitigate concentration risks when trade flows lack geographic diversification.

Market Context

Aircraft Manufacturing forms the apex of the U.S. aerospace industrial base, anchoring complex multi-tier supply chains across propulsion, structures, and avionics. The segment's \$10.0B export portfolio underscores its role as a critical balance-of-trade contributor, but the 60% single-destination share reveals structural exposure uncommon in diversified defense portfolios. This trade dependency likely reflects large commercial aircraft sales to a high-volume buyer (e.g., China or a Middle Eastern hub), creating program-level vulnerabilities if that relationship sours. Workforce fundamentals remain serviceable but not immune to skilled labor shortages plaguing advanced manufacturing nationwide.

Recommended Actions

- **Procurement Directors:** Require Tier-1 aircraft OEMs to disclose end-customer concentration by platform in annual supplier business reviews; flag programs where >40% of revenue derives from the single export destination identified in this analysis.
- **Program Managers:** Diversify platform export campaigns to reduce single-country dependencies; prioritize customer development in allied markets (Europe, Indo-Pacific partners) where regulatory harmonization enables faster certification.
- **Supply Chain Risk Officers:** Model scenario impacts of a 25% demand reduction from the primary export destination; pre-qualify alternate suppliers or domestic absorption capacity to buffer workforce layoffs and preserve critical skills.